Jerry Liu

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Education

University of Waterloo Waterloo, Ontario

BASc, Mechatronics Engineering

- **Awards:** Arthur F. Church Entrance Scholarship ($10,000), awarded to only 1 student in the Mechatronics cohort for outstanding academic and extracurricular achievement; FRC (FIRST Robotics) World Championship qualifier
- **Coursework:** Linear Algebra (MATH 115), Experimental Measurements and Statistical Analysis (MTE 201, in progress), Data Structures and Algorithms (MTE 140), Digital Computation (MTE 121), Calculus 1 and 2

Skills

Languages: Python, TypeScript, SQL, JavaScript, C++
Machine Learning: PyTorch, Transformer Models, Embeddings, OpenCV, YOLOv8, Pose Estimation
LLMs and Agents: LangGraph, ReAct Agents, RAG, Prompt Engineering, LLM Vision Extraction, Gemini API, Claude Code
Backend and Data: FastAPI, NestJS, Node.js, REST APIs, PostgreSQL, PostGIS, Redis/BullMQ, Supabase
Infrastructure: Docker, AWS S3, GitHub Actions (CI/CD), Git, Playwright

Experience

Research Assistant Waterloo, Ontario

University of Waterloo *Sept 2026 – Present*

- Built a physics model of a judo throw in Python that estimates each body part's mass from body measurements and tracks center of mass and 3D joint motion for both athletes, checked against YOLOv8 pose tracking on 10 videos
- Synced two camera angles of judo throws to within 1 frame (at 30 fps) across 12 clips by automatically detecting the moment of the throw from body keypoints, using RTMPose and Pose2Sim
- Fixed uneven video frame rates with FFmpeg and confirmed the sync on a second moment in each throw with 0 frames of error, using OpenCV skeleton overlays to check results visually

Full-Stack Software Engineering Intern (member of the founding team) Austin, TX

Parcyl | Link | Site intelligence platform for real estate diligence; 2-person engineering team *May 2026 – Aug 2026*

- Built a LangGraph multi-agent loop that drives the product as a synthetic user and proposes, builds and live-verifies each fix into a draft PR behind a human merge gate, surfacing 85 verified defects across 6 root causes
- Designed and built an agentic report engine: 11 ReAct agents generating research reports for any property in a 65M-record warehouse, each personalized to the user's portfolio, uploaded documents and prior deals
- Shipped an async document pipeline on S3 and BullMQ that extracts structured facts from PDFs, spreadsheets and scans and feeds them into the chat and report agents, which must cite each fact to its source quote and page
- Architected a feedback-triage agent that deduplicates user reports in three stages and posts a batched Slack digest carrying the PostHog session replay behind every finding, collapsing 50+ raw reports into 5 distinct findings

Research Assistant Waterloo, Ontario

University of Waterloo *August 2024 – June 2025*

- Selected multi-modal sensors for CNC tool-wear monitoring under Dr. Eugene Li and built the Python sensor-to-model pipeline turning high-frequency accelerometer data into PyTorch tensors across 1.5M+ samples
- Prototyped a PyTorch Transformer for remaining-useful-life prediction from the processed time-series data
- Trained and evaluated an NCA+KNN classifier on the same dataset, reaching 79.6% test accuracy

Co-Founder, Logistics Lead Waterloo, Ontario

NeoDev League | Link *May 2024 – June 2025*

- Secured $10,000+ in sponsorships and backed by Convictional (YC W19); reached 150 students across 10 schools

Projects

Neural Chess Engine | Link Nov 2025

- Trained a 6-layer PyTorch neural network on millions of high-ELO games with a 12-channel (12×8×8) encoding
- Implemented a minimax engine with alpha-beta pruning and deployed it to Lichess via the lichess-bot API

Canopi: Rental Matchmaker | Link March 2026

- Implemented a custom compatibility algorithm to score user-to-property matches and integrated the Gemini API to power a natural-language AI chat interface for surfacing user preferences
- Built a full-stack rental platform with Next.js, TypeScript, and Supabase, integrating a custom scraper to aggregate real-time housing data across Canada